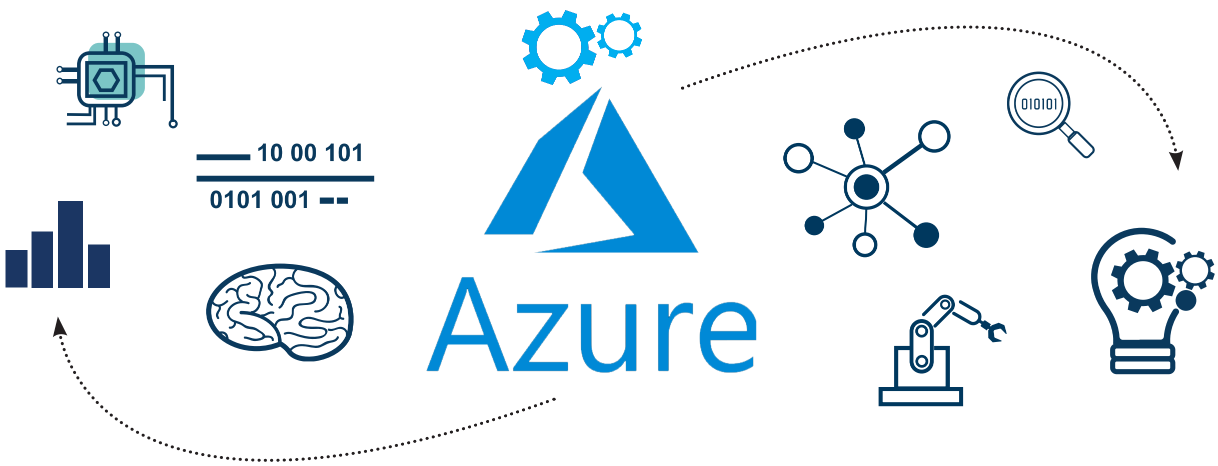
The AI and ML Services that Microsoft Azure Offers

Humongous amounts of data are being generated each moment today. Artificial intelligence and machine learning services are needed to handle and analyse this Big Data. This is where the AI and ML services of Microsoft Azure come in. Read on to find out how they are helping developers and businesses today.



Artificial intelligence (AI) is the traditional model devised to mimic human brain functions by computers using a technology-driven approach along with mathematical and statistical models (e.g., probability). Machine learning (ML) was conceived later to make programming models of brain functions. It aimed to improve the behaviour of AI models using various learning approaches and storing the results of machine learning models for future predictions. This was inspired by the neural network — the core storage layer of the human brain that stores data using the input layer, the hidden layer and the output layer.

There are many popular machine learning frameworks available in the market, many of which are open source. To choose a machine learning tool or framework, one has to apply the problem scenario, data requirement, data volume to process, training data availability, trainability, algorithm support, visualisation and reporting facility, capability to auto-build the knowledge or training model, processing speed required (CPU/GPU), to name a few.

In ML, building the model is a time consuming task as a lot of training is required to get accuracy for the given problem. For example, for text recognition from images, a lot of training needs to be done in order to build the neural network of data sets for better recognition accuracy. This requires higher compute resources and hence greater scalability. This can be well achieved with cloud based platforms and, hence, machine learning services in cloud platforms are very popular today.

Azure machine learning services, including ML Studio, are being widely used for various business scenarios like anomaly detection, sentiment analysis for marketing research, and predictive analysis for customer relationship management, to name a few. But there are more practical business use cases now being experimented using Azure ML services in the banking industry. These include biometric identification, cheque processing using biometric analysis like face detection, and signature detection using pattern matching.